

Project: Customer Purchase Behavior (KNN)

Objective: Build a model to classify customers into purchase categories.

Method: K-Nearest Neighbors (KNN) Algorithm.

Created by: AI Agent (Antigravity)

Problem Statement

Problem: Need to automate customer segmentation.

Context: 5000 records, 5 categories (Electronics to Sports).

Challenge: Significant class imbalance and missing data.

Real-World Use Case

Scenario: Personalized homepage recommendations.

Application: Recommend 'Running Shoes' to 'Sports' users.

Impact: Increased revenue through targeted marketing.

Input Data

Demographics: Age, Income, Account Age.

Behavior: Spending, Session Duration, Page Views.

Categorical: Device Type, Membership Tier.

Target: Purchase Category (0-4).

Concepts Used

1. Imputation: Filling missing values (Mean).
2. Encoding: Converting text to numbers (OneHot).
3. Scaling: Normalizing features (StandardScaler).
4. KNN Algorithm: Distance-based classification.

Step-by-Step Flow

Step 1: Load Data.

Step 2: Clean Data (Impute & Encode).

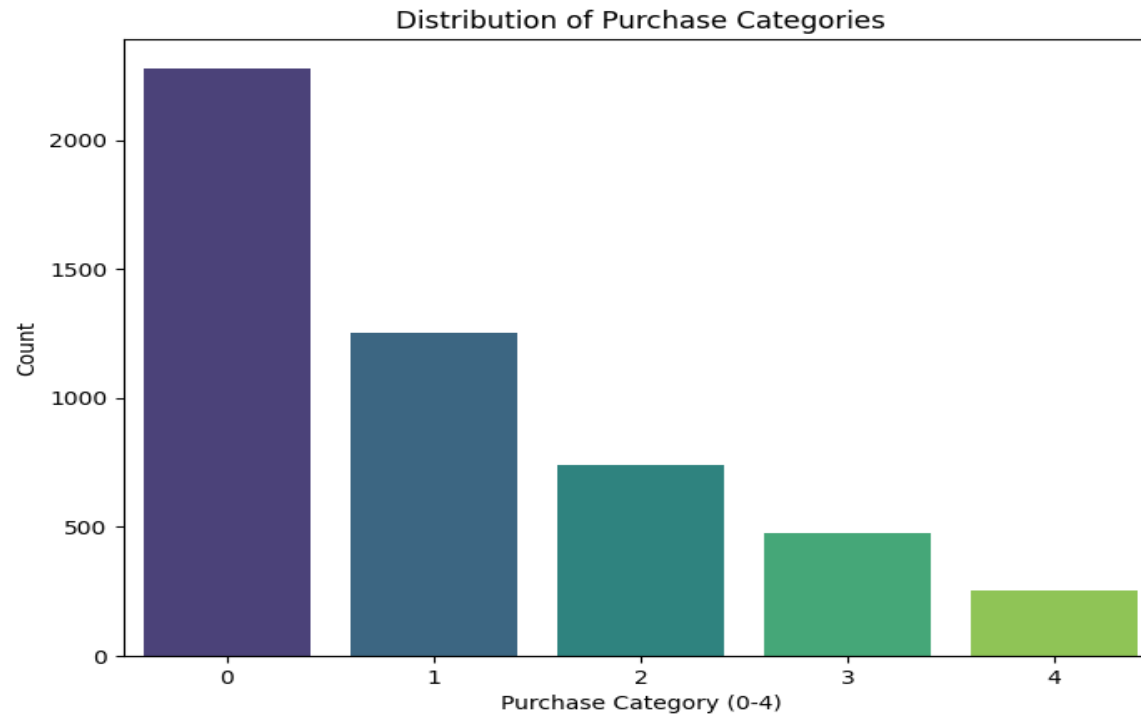
Step 3: Scale Features (Crucial for KNN).

Step 4: Train KNN ($k=5$).

Step 5: Evaluate on Test Set.

Data Visualization: Class Distribution

Note the imbalance towards Class 0 (Electronics).



Code Logic Summary

Pipeline: Imputer -> Scaler -> Model.

Split: 70% Train, 30% Test (Stratified).

Model: KNeighborsClassifier(n_neighbors=5).

Execution Results

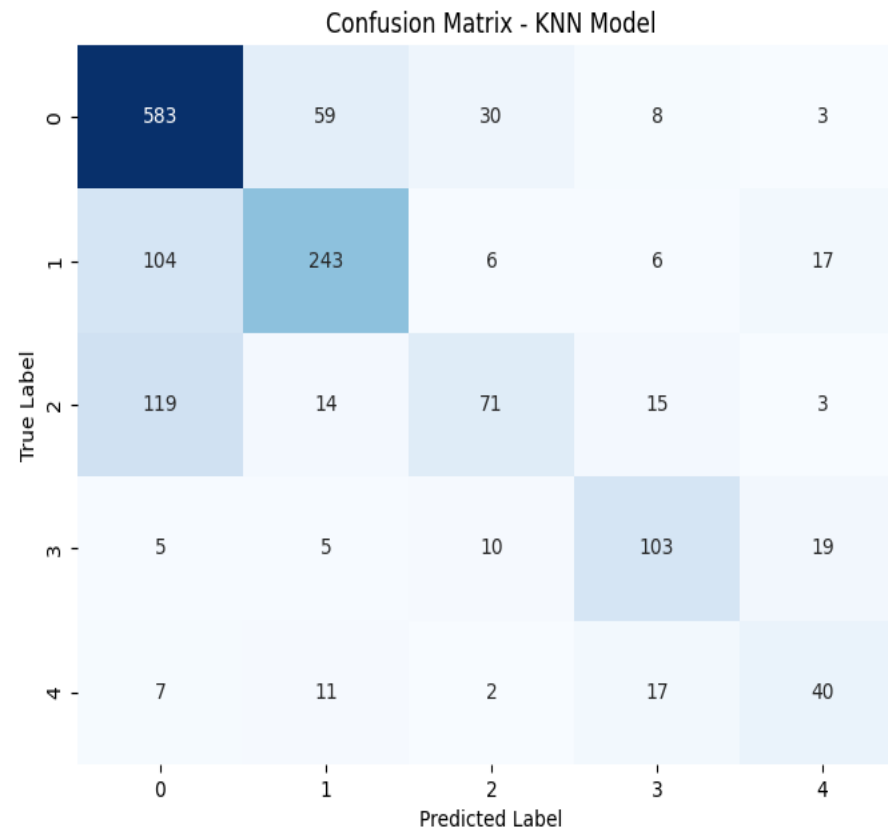
Overall Accuracy: ~69%

Best Class: Electronics (High Recall).

Worst Class: Home & Sports (Low Recall due to less data).

Confusion Matrix

Visualizing where the model makes mistakes.



Observations & Insights

Imbalance Impact: Model favors majority class.

Feature Sensitivity: Scaling was necessary for KNN.

Missing Data: ~1% of rows had missing values, imputation handled this.

Advantages & Limitations (KNN)

Pros: Simple, Intuitive, No training time.

Cons: Slow prediction on large data, Sensitive to outliers.

Interview Corner

Q: Why scale for KNN? A: Distance metrics are sensitive to magnitude.

Q: How to handle imbalance? A: SMOTE, or adjust Class Weights (not available in standard KNN, use distance weights).

Conclusion

We established a baseline with KNN (69%).

Next steps: Try Random Forest to handle non-linearity better.

Recommendation: Collect more data for 'Sports' category.